

整数集 $\mathbb{Z}$ 的构造

定义 ($\mathbb{Z}_{\geq 0}^2$) 记 $\mathbb{Z}_{\geq 0}^2 := \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}$.

Lemma: 定义 $\mathbb{Z}_{\geq 0}^2$ 上的二元关系为: 对 $\forall (m, n), (m', n') \in \mathbb{Z}_{\geq 0}^2$,

$$(m, n) \sim (m', n') \Leftrightarrow m + n' = m' + n$$

则有: $(m, n) \sim (m', n')$ 是集合 $\mathbb{Z}_{\geq 0}^2$ 上的等价关系.

Proof: 对 $\forall (m, n) \in \mathbb{Z}_{\geq 0}^2$, $\because m + n = m + n$

$\therefore (m, n) \sim (m, n)$ $\therefore$ 反身性成立.

对 $\forall (m, n), (m', n') \in \mathbb{Z}_{\geq 0}^2$, 若 $(m, n) \sim (m', n')$, 则有: $m + n' = m' + n$

$\therefore m' + n = m + n'$ $\therefore (m', n') \sim (m, n)$ $\therefore$ 对称性成立.

对 $\forall (m, n), (m', n'), (m'', n'') \in \mathbb{Z}_{\geq 0}^2$, 若 $(m, n) \sim (m', n')$ 且 $(m', n') \sim (m'', n'')$,

则有: $\because (m, n) \sim (m', n')$ $\therefore m + n' = m' + n$

$\because (m', n') \sim (m'', n'')$ $\therefore m' + n'' = m'' + n'$

$$\therefore m + n' + m' + n'' = m' + n + m'' + n'$$

$$\therefore (m + n'') + (n' + m') = (n + m'') + (n' + m')$$

$$\therefore m + n'' = n + m'' \quad \therefore m + n'' = m'' + n$$

$\therefore (m, n) \sim (m'', n'')$ $\therefore$ 传递性成立.

$\therefore \sim$ 是集合 $\mathbb{Z}_{\geq 0}^2$ 上的等价关系. □

定义(整数集 $\mathbb{Z}$) 定义整数集 $\mathbb{Z}$ 为 $\mathbb{Z} := \mathbb{Z}_{\geq 0}^2 / \sim$. 记 $\mathbb{Z}_{\geq 0}^2$ 中包含 (m, n) 的等价类为 $[[m, n]]$.

Remark: 对 $\forall (m, n) \in \mathbb{Z}_{\geq 0}^2$ (其中 $m \in \mathbb{Z}_{\geq 0}, n \in \mathbb{Z}_{\geq 0}$), $[[m, n]] \subseteq \mathbb{Z}_{\geq 0}^2$.

$$[[m, n]] = \{(x, y) \in \mathbb{Z}_{\geq 0}^2 \mid (x, y) \sim (m, n)\}$$

$$= \{(x, y) \in \mathbb{Z}_{\geq 0}^2 \mid x + n = m + y\}$$

这两个加法“+”是 $\mathbb{Z}_{\geq 0}$ 中的加法

$$\mathbb{Z} := \mathbb{Z}_{\geq 0}^2 / \sim = \{[[m, n]] \mid (m, n) \in \mathbb{Z}_{\geq 0}^2\}$$

Lemma (从 $\mathbb{Z}_{\geq 0}$ 到 $\mathbb{Z}$ 的嵌入映射) 定义映射 $f: \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}$ 为:

$$f: \mathbb{Z}_{\geq 0} \longrightarrow \mathbb{Z}$$

$$m \longmapsto [[m, 0]]$$

则有: 映射 f 是单射.

Proof: 对 $\forall m \in \mathbb{Z}_{\geq 0}$, $\because m \in \mathbb{Z}_{\geq 0}$ 且 $0 \in \mathbb{Z}_{\geq 0} \therefore (m, 0) \in \mathbb{Z}_{\geq 0}^2$

$$\therefore f(m) = [[m, 0]] \in \mathbb{Z} \quad \therefore f(\mathbb{Z}_{\geq 0}) \subseteq \mathbb{Z}$$

对 $\forall m, n \in \mathbb{Z}_{\geq 0}$, 有:

$$\text{若 } m = n, \text{ 则有: } (m, 0) = (n, 0) \in \mathbb{Z}_{\geq 0}^2 \quad \therefore [[m, 0]] = [[n, 0]]$$

$$\therefore f(m) = [[m, 0]] = [[n, 0]] = f(n) \quad \therefore f \text{ 是映射.}$$

若 $f(m) = f(n)$, 则有: $[[m, 0]] = [[n, 0]]$

$$\therefore (m, 0) \in [[m, 0]] = [[n, 0]] \quad \therefore (m, 0) \sim (n, 0)$$

$$\therefore m+0 = n+0 \quad \therefore m=n \quad \therefore f \text{ 是单射.} \quad \square$$

Remark: 今后将不加说明地将 $\mathbb{Z}_{\geq 0}$ 视同 $\mathbb{Z}$ 的子集.

定义 ($\mathbb{Z}$ 中的加法和乘法) 对 $\forall [[m, n]], [[r, s]] \in \mathbb{Z}$, 定义它们的和与积为:

$$[[m, n]] + [[r, s]] := [[m+r, n+s]]$$

$$[[m, n]] \cdot [[r, s]] := [[mr+ns, nr+ms]]$$

Lemma: $\mathbb{Z}$ 中的加法运算良定义.

Proof: 对 $\forall [[m, n]], [[m', n']], [[r, s]], [[r', s']] \in \mathbb{Z}$,

若 $[[m, n]] = [[m', n']]$ 且 $[[r, s]] = [[r', s']]$, 则有:

$$\therefore [[m, n]] = [[m', n']] \quad \therefore (m, n) \sim (m', n') \quad \therefore m+n' = m'+n$$

$$\therefore [[r, s]] = [[r', s']] \quad \therefore (r, s) \sim (r', s') \quad \therefore r+s' = r'+s$$

$$\therefore (m+n') + (r+s') = (m'+n) + (r'+s)$$

$$\therefore (m+r) + (n'+s') = (m'+r') + (n+s)$$

$$\therefore (m+r, n+s) \sim (m'+r', n'+s') \quad \therefore [[m+r, n+s]] = [[m'+r', n'+s']]$$

$$\therefore [[m, n]] + [[r, s]] = [[m+r, n+s]] = [[m'+r', n'+s']] = [[m', n']] + [[r', s']]$$

$\therefore \mathbb{Z}$ 中的加法运算良定义. $\square$

Lemma: $\mathbb{Z}$ 中的乘法运算良定义.

Proof: 对 $\forall [m, n], [m', n'], [r, s], [r', s'] \in \mathbb{Z}$,

若 $[m, n] = [m', n']$ 且 $[r, s] = [r', s']$, 则有:

$$\therefore [m, n] = [m', n'] \quad \therefore (m, n) \sim (m', n') \quad \therefore m + n' = m' + n$$

$$\therefore [r, s] = [r', s'] \quad \therefore (r, s) \sim (r', s') \quad \therefore r + s' = r' + s$$

$$\therefore m + n' = m' + n \quad \therefore (m + n')r = (m' + n)r$$

$$\therefore n + m' = n' + m \quad \therefore (n + m')s = (n' + m)s$$

$$\therefore (m + n')r + (n + m')s = (m' + n)r + (n' + m)s$$

$$\therefore mr + n'r + ns + m's = m'r + nr + n's + ms$$

$$\therefore (mr + ns) + (n'r + m's) = (m'r + n's) + (nr + ms)$$

$$\therefore (mr + ns, nr + ms) \sim (m'r + n's, n'r + m's)$$

$$\therefore [mr + ns, nr + ms] = [m'r + n's, n'r + m's]$$

$$\therefore r + s' = r' + s \quad \therefore m'(r + s') = m'(r' + s)$$

$$\therefore s + r' = s' + r \quad \therefore n'(s + r') = n'(s' + r)$$

$$\therefore m'(r + s') + n'(s + r') = m'(r' + s) + n'(s' + r)$$

$$\therefore m'r + m's' + n's + n'r' = m'r' + m's + n's' + n'r$$

$$\therefore (m'r + n's) + (n'r' + m's') = (m'r' + n's') + (n'r + m's)$$

$$\therefore (m'r + n's, n'r + m's) \sim (m'r' + n's', n'r' + m's')$$

$$\therefore (mr + ns, nr + ms) \sim (m'r' + n's', n'r' + m's')$$

$$\therefore \llbracket mr+ns, nr+ms \rrbracket = \llbracket m'r'+n's', n'r'+m's' \rrbracket$$

$$\therefore \llbracket m, n \rrbracket \cdot \llbracket r, s \rrbracket = \llbracket mr+ns, nr+ms \rrbracket$$

$$= \llbracket m'r'+n's', n'r'+m's' \rrbracket = \llbracket m', n' \rrbracket \cdot \llbracket r', s' \rrbracket$$

$\therefore \mathbb{Z}$ 中的乘法运算良定义 $\square$